

Where the Seconds Go:

A Speed Limit and an Accounting for Mixture-of-Experts Decode on One Consumer GPU

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Abstract

A Mixture-of-Experts (MoE) model too large for one consumer GPU keeps most of its experts in host memory, and every decoded token waits while some of them are read. We ask how fast such a machine could decode, how far real systems are from that, and what separates them from it. **A host-memory law.** Time per token is a fixed GPU term plus the expert bytes read from host memory over the rate of the read paths in use. Written on each machine from its own bandwidth probe before any model run, it predicted 29 configurations on 7 machines with 3.7% median error, and under leave-one-host-out validation its max-of-paths form beats the additive alternative (3.1% against 4.5% median error). **A speed limit.** Belady’s optimum with bypass on the models’ own routing, with two host read paths and the machine’s measured rates, bounds any exact-routing system at a given expert budget. Published trace-scored offloading measurements reach a median 9.5% of their own limit; on one machine, stock llama.cpp reaches 9–20%, FreeToken 20–40%, and our cache 25–42% (33–57% of a limit tightened in four ways). **An accounting.** Shapley attribution over every order of fixes says where the rest goes: missing foresight is the largest cost where the host binds (37–53% of the gap), host reads serialised with GPU work where it does not (39–46%); GPU kernel efficiency, replacement policy and host work are never largest in any order. **What the traces say.** Across nine models, the best online policy reads 35–110% more than the optimum; four tokens of foresight beat all of them; static placement reads 1.4–13.2× the optimum because the hot set turns over 33–75% every 1,000 tokens; and batched speculative verification recovers a median 15–32% of reads at best, so speculation is not foresight. **A system.** An expert cache in llama.cpp that acts on the fixable items decodes gpt-oss-120b 15–29% and Qwen3-30B-A3B 3–15% faster than FreeToken at equal GPU memory and 2.0–3.8× as fast as llama.cpp, 1.51–1.58× FreeToken behind a link of half the bandwidth, with the gain attributed by ablation. Every prediction was committed before the measurement it predicts; 97 clauses are scored in the appendix.

1 Introduction

Open-weight Mixture-of-Experts models now carry 30–120 B parameters and activate a few billion per token; a desktop GPU has 24–32 GB. Running such a model at batch 1 therefore means keeping most experts in host DRAM, and each decoded token then reads its routed experts from one of three places: GPU memory, host memory by the CPU that runs the expert, or host memory across PCIe into a GPU slot [16, 12]. A line of systems has built on this: GPU expert caches with demand fetch [8, 41, 34], CPU execution of misses [18, 5, 51, 14, 50, 53], permiss choices between the two [44], and expert prediction [15, 52, 39]. Each reports a speed-up over a baseline of its choosing. None states how fast its machine *could* decode the model within the same GPU memory, or how much of the remaining time any design could remove, and the trace-level evaluation contracts proposed for this field [49] count cache misses, not seconds.

We answer those questions on consumer hardware (RTX 5090 hosts with desktop CPUs, rented by the hour) with two models, gpt-oss-120b [28] and Qwen3-30B-A3B [43], routing traces of nine, and 52 published measurements from 13 sources. The method is Hoeffler and Belli’s [13]: a performance model with bounds, statistically sound comparisons, and every prediction committed to a public repository before the machine it concerns was started (Appendices A and B).

Contributions.

- **A host-memory law, validated as a predictor** (Section 3): time per token = $G + \max(X_c/B_c, X_p/B_p, (X_c+X_p)/B_{cp})$, host bytes X from the engine’s counters and rates B from a probe. Predicted blind on 7 machines, its median error is 3.7%. The max-of-paths form is FreeToken’s per-layer split [44] lifted to a token; what is new is that it is tested as a predictor, against the additive and per-layer alternatives, and that each of its failures is traced to a measured cause.
- **A speed limit in seconds** (Section 4): a bound on any exact-routing system with C expert slots per layer, from the exact Belady optimum with bypass, two host read paths and measured ceilings, reported

beside every measured speed (Table 1) and tightened four ways (Table 3). Cache-level oracles exist [49]; a limit in seconds with two paths, compared against real systems, does not.

- **An accounting** (Section 4): Shapley attribution of the gap between each system and the limit, which no ordering of fixes can bias, with the regime it predicts: foresight where the host binds, overlap where the GPU does.
- **What the traces say** (Section 6): across nine models, which policy earns its reads, how fast the hot set drifts, how many tokens of foresight are worth half the optimum ($W_{50} \approx 0.59 (C/k)^{1.33}$, with intervals), and what batched speculative verification cannot buy.
- **An audit** (Section 7): 52 published measurements placed against their own limit.
- **A system as the instrument** (Section 5): a llama.cpp expert cache whose per-machine miss split comes from the law. It leads FreeToken at every budget on two models, ties it on a third, runs $2.0\text{--}3.8\times$ llama.cpp, and an ablation says which parts matter.

What we do not claim. Running missed experts on the CPU [18, 5], caching hot experts with CPU-executed misses [51, 14, 21], splitting a layer’s misses between CPU and PCIe from a bandwidth model [44, 53], a PCIe turnover ceiling [47], bytes over bandwidth as a bound [46], Belady with bypass as a cache-level oracle [49] and a table of systems against the all-in-VRAM speed [39] are all published. Our system uses the first three; what is new is the limit in seconds, the accounting, the trace results, the audit, and the measurements.

2 Setting and Method

Workload. Single-request decode. gpt-oss-120b (MXFP4 experts; 36 MoE layers of 128 experts, top-4, 13.25 MB per expert) and Qwen3-30B-A3B in BF16 (48 layers of 128 experts, top-8, 9.44 MB per expert). Prompts are the 30 AIME-25 problems in FreeToken’s own benchmark format; each system decodes 256 tokens per problem, greedily, after one warm-up request on a fixed held-out text, in one session in which the cache carries over between problems. One client measures every system with FreeToken’s own metric, (tokens−1) over the time between the first and last token’s arrival.

Machines and probes. RTX 5090 (32 GB, 1.79 TB/s datasheet) hosts with desktop CPUs, rented by the hour. Each job first runs a probe (concur) that reads host memory from t CPU threads, over PCIe with zero-copy kernels and with the copy engine, and from both at once; FreeToken’s own probe (ft bench bw) runs too. We call the CPU rate at the helper thread count B_c , the zero-copy PCIe rate B_p and the combined rate B_{cp} . GPU clocks

cannot be locked on these rentals; we report each card’s power limit. The headline machine (Table 1, jobs 080–082) is one rental of a Ryzen 9 9950X3D host whose card reports a 17001 MHz memory clock, against 14001 MHz on 24 of the 25 other RTX 5090 rentals that recorded one (15501 (1) on the rest), $1.21\times$ the bandwidth the limit assumes; a second rental of the same CPU model (jobs 079, 087) has the common clock. We call them listing B and listing A.

Equal memory. llama.cpp keeps whole layers’ experts on the GPU (`--n-cpu-moe n`), our cache keeps C experts of every layer, and FreeToken takes a cache rate; we match the number of experts on the GPU (within 2.5%) and report measured VRAM. FreeToken preallocates 28.4–30.1 GiB whatever its rate.

Statistics and protocol. Ratios are of mean speeds, paired by problem, with 95% percentile intervals from 10,000 bootstrap resamples; they cover problem sampling within one launch on one machine, not rentals (two rentals of one CPU model differed by 19–22% in host bandwidth). Where a system has variants (our FETCH tables, FreeToken’s offload and hybrid backends), a first launch picks each system’s faster variant per budget and a second launch, which the comparison uses, runs only those (one row of Table 1 has no second launch). Every job’s predictions are in the header of its script, committed and pushed before the machine starts; Appendix B scores every clause under one rule.

3 A Host-Memory Law

With experts partly in host memory, a decode step does two kinds of work: GPU work that every token needs (attention, dense projections, resident experts, the output head, launches), and reads of the experts that are not resident. The second is limited by host memory. The CPU reads X_c bytes per token (the experts it runs), PCIe reads X_p bytes (the experts copied to the GPU, whether on demand or ahead of time), and when both run at once they share the DRAM:

$$T = G + \max\left(\frac{X_c}{B_c}, \frac{X_p}{B_p}, \frac{X_c + X_p}{B_{cp}}\right). \quad (1)$$

X_c and X_p come from the engine’s counters of misses, fetches and admissions, which do not depend on the machine. B_c , B_p and B_{cp} come from the machine’s probe. G is one constant per engine and model, fitted once on one host (4.82 ms for our cache on gpt-oss-120b, 5.06 ms for llama.cpp) and then frozen in the repository. The form is FreeToken’s residual-bandwidth split of one layer [44], lifted to the token; the difference is that the three rates are probed rather than assumed to add, and that the law is tested as a predictor.

Table 1: Decode speed at equal GPU expert memory on one machine (RTX 5090 + Ryzen 9 9950X3D; host memory read by the CPU at 71.6 GB/s and over PCIe at 53.2 GB/s). 30 AIME-25 problems, first 256 decode tokens, greedy, session. Ours uses the FETCH split computed from this machine’s bandwidth probe; FreeToken uses its faster backend per budget. Ratios are of mean speeds, paired by problem, with 95% bootstrap intervals; the comparison launch is separate from the launch that picked each system’s variant. *Speed limit*: the fastest any exact-routing system with the same slots per layer can decode on this machine (Section 4; exact optimum, highest probed host rate, datasheet GPU rate; Table 3 tightens it). *All in VRAM*: stock llama.cpp with every weight on an RTX PRO 6000 (the RTX 5090’s datasheet memory bandwidth, 96 GB). [†]Launch 1 for both systems (job 080): FreeToken’s backend was picked on the launch it is scored on, which favours it, and the row has no confirmation launch.

Model	Experts on GPU	llama.cpp (tok/s)	FreeToken (tok/s)	Ours (tok/s)	Ours ÷ FreeToken (95% CI)	Ours ÷ llama.cpp	Speed limit	Ours, % of limit	All in VRAM
gpt-oss-120b	11%	34.9	54.0 (hybrid)	69.9	1.294 [1.278, 1.312]	2.00×	172	41%	255
gpt-oss-120b	25%	40.0	85.6 (hybrid)	109.2	1.275 [1.254, 1.295]	2.73×	430	25%	255
gpt-oss-120b	40%	47.7	132.2 (offload)	152.5	1.154 [1.134, 1.172]	3.20×	519	29%	255
Qwen3-30B-A3B	12.5%	19.5	38.7 (hybrid)	40.0	1.032 [1.022, 1.042]	2.05×	96	42%	178
Qwen3-30B-A3B	25%	22.1	54.7 (hybrid)	63.1	1.153 [1.134, 1.173] [†]	2.86×	214	30%	178
Qwen3-30B-A3B	43.75%	28.4	103.1 (offload)	108.2	1.049 [1.036, 1.064]	3.81×	308	35%	178

Table 2: The law predicted blind: each machine wrote its prediction from its own bandwidth probe, with constants frozen in the public repository, before any model run. Error of predicted over measured speed, gpt-oss-120b (cache rows) and llama.cpp --n-cpu-moe (last column, %).

Host CPU	Job	n	Median err %	Max err %	llama.cpp err %
Core Ultra 7 270K (8P+16E)	069c	5	1.5	11.9	+119
EPYC 7352 (Zen 2)	069c	5	3.2	11.7	+10
EPYC 9655 (12-ch. DDR5)	069c	5	18.7	31.9	+16
Ryzen 9 7900	072	5	4.9	8.2	+4
Ryzen 9 9950X3D (A)	073	8	1.6	4.5	–
Ryzen 9 7950X	074	1	4.0	4.0	–
Ryzen 9 9950X	076	4	2.5	5.0	–
Ryzen 9 9950X3D (B)	082	0	–	–	+4, +4

Blind test. On every later machine the job wrote the law’s prediction from its own probe before any model ran (Table 2). Over 29 configurations of our cache on 7 machines (33 measurements; one machine ran twice) the median error is 3.7%; without the 12-channel EPYC 9655 it is 3.3% over 24 configurations (90th percentile 7.5%). llama.cpp --n-cpu-moe is predicted within 4% on the Ryzen hosts, including two Qwen3 configurations predicted on the headline machine (+3.6%, +4.0%; for Qwen3, $G=4.3$ ms, the mean residual of three llama.cpp runs on another host).

Is the max the right form? In 19 of the 29 configurations nothing is copied, so they test one roof. We therefore refit the law and its alternatives on the same 33 measurements under leave-one-host-out validation (each of the 7 hosts predicted with constants fitted on the other six): the token-level max of Eq. (1) with G refit has a median error of 3.1% (90th percentile 9.4%); the additive form $G + X_c/B_c + X_p/B_p$, 4.5% (11.3%); a per-layer max summed over layers from the engine’s per-layer miss his-

tograms, 3.2% (10.2%). The additive form ties in sample but misses the three rows that copy misses on demand behind a 53 GB/s link, where both paths carry most of the bytes, by -11%, while the max is within $\pm 0.2\%$ there; on the slow-link copy rows the additive form is the closer of the two. The per-layer form neither beats the max nor explains its failures: the Jensen gap between a sum of per-layer maxima and the maximum of sums is a fifth of the Core Ultra’s residual, none of the EPYC 7352’s, and has the wrong sign on the EPYC 9655 and on the prefetch rows. We keep Eq. (1).

Where it fails, and why. Each failure now has a measured cause. (i) Copying misses on demand (FETCH) at 25% is over-predicted by +11.7% on the Core Ultra 7 and +11.4% on the EPYC 7352 (links of 36 and 26 GB/s); +8.8% and +8.5% with G refit. The engine’s per-layer timers show why: a CPU miss served beside a copy takes 20–62% longer than one served alone, so the two paths share DRAM less evenly than the probe’s combined rate says; a variant of the law that uses the probe’s per-path rates under contention and a per-layer hand-off constant predicts both rows within +3.1% and +4.1%, at a worse median (4.2%). (ii) llama.cpp on a Core Ultra 7 with 8 performance and 16 efficiency cores runs at 46% of the law at its default thread count, because it splits every matrix evenly across its threads; our helpers claim work dynamically and stay on the law on the same machine. We did not test llama.cpp restricted to the performance cores. (iii) On a 12-channel EPYC 9655 (559 GB/s) host memory stops being the limit (+16 to +29%): the per-layer timers show a floor of 55–70 μ s per layer with CPU misses that no desktop host has (their intercepts are at or below zero), the latency of the hand-off at a bandwidth where a 13 MB expert takes 24 μ s. (iv) Prefetch configurations are under-predicted by -8.7 to -3.4%: their copies overlap attention, which the law charges in full.

What the law says. Every mechanism in this design space acts only through the bytes read from host memory per token and the number of read paths in use. A second path adds bandwidth on every host we measured but the 12-channel EPYC (B_{cp} against $\max(B_c, B_p)$): 8–17% on the Ryzen hosts, 1% on the Core Ultra, 33% on the Zen 2 EPYC), and that is the whole gain available to fetching, prefetching or splitting misses between the CPU and PCIe. Prefetching never reduces the bytes. G (5.17ms refit) exceeds the whole all-in-VRAM token (3.92ms) because it absorbs a shortfall the probe does not see: the helpers read at $1.09\text{--}1.31\times$ the probe’s time per expert on desktop hosts, and a fitted rate factor is collinear with G at this sample size. And the “host lottery” is the DRAM: two rentals of the same CPU differed by 19–22% in speed, which is 62 against 47 GB/s of host bandwidth.

4 The Speed Limit, and Where the Seconds Go

Bound. Consider any system that executes the model’s exact routing with at most C experts of each layer resident on the GPU. Per token, each of the Lk routed experts is either served by a resident copy (the GPU reads it from VRAM) or not, in which case its S bytes are read from host memory once more: by the CPU that runs it, or by the copy that brings it into a slot. Let R^* be the reads per token of Belady’s MIN with bypass [2, 9, 48] applied per layer to the routing trace, the fewest reads of any policy, even one that knows the future (an expert may be evicted after serving the current token; a policy that keeps every requested expert until the token ends reads 0.5–4.0% more). If c experts per token run on the CPU, host memory supplies at least $\max(R^*, c)S$ bytes and the GPU reads the dense bytes D plus $(Lk - c)S$. With everything overlapped,

$$\bar{T} \geq \min_{0 \leq c \leq Lk} \max\left(\frac{D + (Lk - c)S}{B_{\text{gpu}}}, \frac{\max(R^*, c)S}{B_{\text{host}}}\right), \quad (2)$$

with B_{gpu} the GPU’s datasheet rate and B_{host} the highest host read rate the probe measured by any method (87.5 GB/s on the headline machine). The bound assumes that an expert run on the CPU is read from DRAM (the optimum’s shortest re-read distance is 0.5 GB of host traffic, far beyond any CPU cache), that computation runs at memory speed, and that the two read paths share B_{host} freely; it drops all latencies and starts from an empty cache. B_{host} is the highest rate our probe reached, not a hardware peak, so a system that reads host memory faster than the probe could exceed the limit (a second rental of the same CPU probed 90.3 GB/s). It can exceed the all-in-VRAM speed, because the CPU’s bandwidth adds to the GPU’s. The traces are the models’ own greedy text on the 30 AIME-25 prompts, templated and tokenised exactly as the server does, with routing recorded teacher-forced by our llama.cpp instrumentation; each system’s

Table 3: The speed limit under each tightening, tokens per second on the headline machine, and our cache as a percentage of the limit as published (exact optimum, highest probed host rate, datasheet GPU rate) and with every tightening at once. *084*: the optimum used in the earlier draft; *exact*: Belady’s MIN with bypass that may evict an expert after serving it; *pool*: one pool of LC slots shared across layers; *median*: B_{host} the median of six concurrent probe samples instead of the highest; *GPU*: the GPU term from the measured all-in-VRAM time instead of the datasheet rate; *layers*: the per-layer sum, which no system without cross-layer prefetch can beat; *all*: exact, pool, median and GPU together (a warm start, not included, changes the limit by at most 1.4%).

Model	Experts on GPU	084	exact	pool	median	GPU	layers	all	Ours, % of exact	% of all
gpt-oss-120b	11%	169	172	182	159	172	163	168	41	41
gpt-oss-120b	25%	427	430	474	397	279	365	278	25	39
gpt-oss-120b	40%	519	519	519	517	279	500	278	29	55
Qwen3-30B-A3B	12.5%	92	96	101	89	96	93	93	42	43
Qwen3-30B-A3B	25%	209	214	234	197	192	193	191	30	33
Qwen3-30B-A3B	43.75%	308	308	308	307	192	298	191	35	57

own greedy text drifts from it within the first few hundred characters on gpt-oss (Section 5).

Tightening it. Table 3 changes one assumption at a time. A single pool of LC slots shared across layers, the true equal-memory constraint, needs 4.5–18.6% fewer reads than per-layer slots and raises the four host-bound limits by 5–10%; it leaves the two largest budgets unchanged, where the limit is the machine’s aggregate bandwidth and read counts no longer matter. Starting the optimum warm, after 1024 tokens, changes it by at most 1.4%. B_{host} is one probe sample; the six concurrent samples on this machine read 80.7 GB/s at the median (mean 81.1, 95% interval 78.4–84.3), and the median lowers the host-bound limits by 8%. The GPU term is the one that moves most: at batch 1 the all-in-VRAM runs reach only 52% (gpt-oss) and 61% (Qwen3) of the datasheet rate, and with the measured rate the limit at 25% and above collapses to one aggregate line, 279 and 192 tok/s. A per-layer sum, $\sum_l \max(\text{GPU}_l, \text{host}_l)$ with the optimum’s reads per layer, which no system without cross-layer prefetch can beat, sits 1–5% of a measured token below the token-level maximum (3–11% if dense work is also serialised). With every tightening at once, our cache stands at 33–57% of the limit, FreeToken at 29–54% and stock llama.cpp at 12–21%; against the limit as published, 25–42%, 20–40% and 9–20%. Even with every weight in VRAM, our cache with every expert in its slots matches or beats stock llama.cpp ($0.997\times$ and $1.047\times$): the cache costs nothing when the model fits. The traces represent the measured runs: replayed through a simulator of our deployed policy, they give hit rates within 1.4 points of the engine’s own counters on gpt-oss and within 0.6–5.3 points on Qwen3.

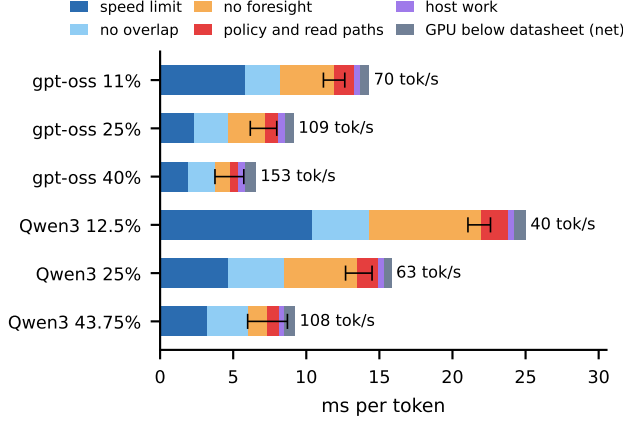


Figure 1: Where the seconds go: Shapley attribution of the gap between our cache’s measured time per token and the speed limit, for every cell of Table 1, averaged over all 120 orders in which the five fixes can be applied. Fixes: overlapping host reads with GPU work; the optimum’s reads instead of the best online policy’s (foresight); the best online policy’s reads at the probe’s highest rate instead of the engine’s reads through Eq. (1) (policy and read paths); the engine’s per-token host work; and GPU kernels at the datasheet rate. Whiskers: the range of foresight’s marginal cost over the orders.

The accounting. An earlier draft added one cost at a time to the limit in a fixed order and found foresight largest in every cell; costed after GPU efficiency instead, it is smaller than that in two cells, and costed after overlap it vanishes at the largest budgets. Shapley values remove the choice: each fix’s share is its average marginal cost over every order (Fig. 1), a method with a precedent in performance debugging of microservices [23]. The result has two regimes, and Eq. (2) says which one a cell is in. In the four cells where the limit binds on host reads ($c^*=0$: gpt-oss at 11 and 25%, Qwen3 at 12.5 and 25%), missing foresight is the largest cost, 37–53% of the gap (26–31% of the token); in the two where it binds on the GPU (gpt-oss 40%, Qwen3 43.75%, balanced by running 13 and 30 experts per token on the CPU), host reads serialised with GPU work is largest, 39–46% of the gap (28–30% of the token), and foresight second at 23%. Policy and read paths cost 12–16% of the gap, of which most is the difference between the probe’s highest rate and the 77.7 GB/s the engine’s own law uses (at that rate the term falls to 0–10%); per-token host work 3–10%; and GPU kernels below datasheet, net of the overlap the engine achieves, 4–16%. Under this accounting, in every order of every cell the largest cost is foresight or overlap; GPU efficiency, policy and host work never are. Charging the all-in-VRAM shortfall in full instead raises GPU efficiency to 26–28% of the gap in the GPU-bound cells, where it is then the largest cost in 40 and 15 of the 120 orders,

and changes no host-bound cell. Foresight is largest in every one of the 120 orders at Qwen3 12.5%, in 115 of them at gpt-oss 11% and Qwen3 25%, and ties overlap within 0.14 ms at gpt-oss 25%. Foresight is the one term no engineering of read paths can recover (Section 6).

One token, profiled. On Qwen3 at 43.75% with our earlier fixed FETCH table (listing A, job 079), Nsight Systems splits a token of our cache (FreeToken) into 4.93 (5.20) ms of expert copies over PCIe, 2.29 (2.21) of resident-expert products, 1.65 (1.34) of attention projections, 0.49 (0.37) of attention, 0.38 of output head and 0.97 (0.40) of norms, small kernels and cache control; our host-side time is 0.27 ms. In that configuration neither system overlaps its PCIe copies with GPU compute; doing so would save 5–10%. Our extra 0.5 ms of GPU work comes from unfused batch-1 kernels: we launch $1.6\times$ as many.

5 The Expert Cache

Design. Our cache is a 3,700-line patch to llama.cpp. Each MoE layer has C GPU slots. Routing is decided on the GPU; resident experts run there, and each miss either runs on CPU helper threads, which the GPU signals through a mailbox in mapped memory, or is copied into a slot and run on the GPU (FETCH). Admission follows a decayed-frequency score (half-life 16 tokens, an admission margin $\kappa=1$; a llama.cpp pull request admits by a decayed heat with hysteresis in the same spirit [26]) and copies in the background. Sampling runs on the GPU. The cache serves single-token steps; prompts and batches above one run llama.cpp’s stock graph with every expert in host memory. The one per-machine decision is the FETCH table $f(n)$: of a layer’s n missed experts, how many to copy. A per-layer form of Eq. (1) gives it,

$$T(n, f) = \max\left(\frac{fS}{B_p} + g, [n > f](L_c + \frac{(n-f)S}{B_c}), [0 < f < n] \frac{nS}{B_{cp}}\right), \quad (3)$$

where g is the GPU’s expert work per layer ($48 \mu\text{s}$ for Qwen3, $37 \mu\text{s}$ for gpt-oss, from profiles) and $L_c=20 \mu\text{s}$ the hand-off; the table picks the f with the smallest T and is computed from the probe when the job starts.

Where the gain comes from. Figure 2 builds the system up from stock llama.cpp on one machine. Placing each layer’s hot experts is the first gain: a static cache of the 32 experts per layer most requested on other text (the models’ own MATH-500 outputs) runs $1.31\times$ and $1.89\times$ stock, which pins whole layers instead. Its hit rate (52% and 67%) depends on how well the profile matches the workload: profiled in hindsight on the measured problems it hits 75% and 72% and runs faster (hatched). Replacing experts online closes that gap without a profile, if admissions are rationed: LRU admits every miss, and on Qwen3 its copies raise host reads from 127 to 188 experts per token, so it runs $0.88\times$ the static cache ($1.30\times$ on

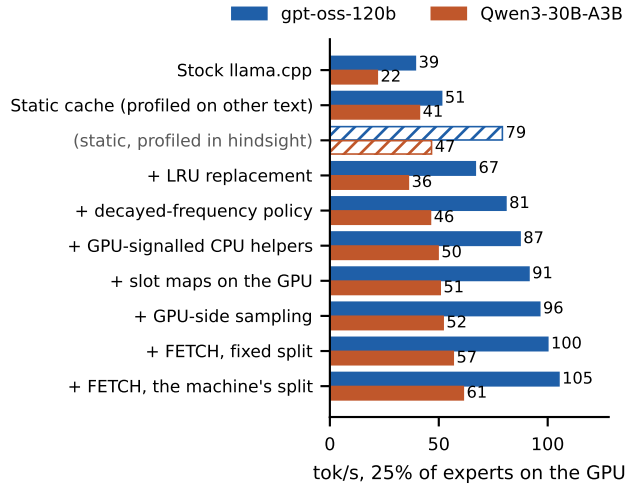


Figure 2: Ablation at 25% of experts on the GPU (15 AIME-25 problems, 256 tokens, listing A; job 087). Each step adds one component to the one above; the hatched row is not a step: a static cache of each layer’s 32 most requested experts counted on the measured problems themselves.

gpt-oss); the decayed-frequency policy reads 41 and 118 per token, runs $1.58\times$ and $1.12\times$ the profiled static cache, and the hindsight cache runs 0.98 [$0.95, 1.01$] and 1.01 [$0.98, 1.04$] times as fast as it. On traces LRU misses only 2–11% more than decayed frequency at these budgets (Section 6); the rest of its cost here is the copies it admits. The GPU-signalled CPU helpers, GPU slot maps, GPU sampling and FETCH add 2–9% each, and the law’s FETCH table 5–8% over the fixed one. The full system runs $2.68\times$ (gpt-oss) and $2.81\times$ (Qwen3) stock llama.cpp. (An earlier run of this ladder, job 082, had a “static” step that loaded no experts; with nothing resident the cache runs $0.74\times$ and $0.75\times$ stock here, $0.72\times$ and $0.74\times$ in job 082.)

Head to head. Table 1 compares the three systems at six budgets on listing B. Ours leads FreeToken at every budget, by 15–29% on gpt-oss and 3–15% on Qwen3, and runs 2.0 – $3.8\times$ llama.cpp. The lead is largest when many experts miss and the CPU path carries them (gpt-oss at 11–25%). It is smallest at Qwen3 12.5%, where our cache already reads host memory at 62 GB/s, 80% of the machine’s combined rate, and at Qwen3 43.75%, where few experts miss and FreeToken’s fused GPU kernels count. The fixed-table ratio at gpt-oss 25% is 1.217, 1.211 and 1.214 on three rentals (jobs 076, 077, 081), so the ratios do not depend on listing B’s faster memory, though its absolute speeds do. On FreeToken’s own headline model, Qwen3.6-35B-A3B in BF16 (256 experts of 6.3 MB per layer, hybrid linear attention), the two systems are close: ours runs 1.030 [$1.023, 1.038$] times FreeToken at

Table 4: FETCH tables at 43.75% on Qwen3-30B-A3B (launch 1): entry n is how many of a layer’s n missed experts are copied over PCIe; the rest run on the CPU. Ratio to the fixed table, paired, 95% CI.

Table	$f(1..8)$	tok/s	vs fixed
Lite	0,0,1,1,1,2,2,3,3	109.0	1.057 [1.052, 1.062]
Law	0,0,1,1,2,3,3,4,5	108.0	1.048 [1.041, 1.055]
Law, fetch a single miss	0,1,1,1,2,3,3,4,5	104.1	1.010 [1.007, 1.012]
Fixed	0,1,1,2,3,3,4,5,6	103.1	1
None (CPU only)	0,...,0	100.7	0.977 [0.970, 0.984]
All (fetch every miss)	0,1,...,8	98.7	0.957 [0.953, 0.962]

12.5% and ties at 25 and 37.5% (1.010 [$0.998, 1.021$], 1.003 [$0.994, 1.013$]), at 1.98 – $2.48\times$ llama.cpp (job 086, listing A). FreeToken as shipped, with its automatic backend and cache size, runs this model at 93.8 tok/s on that machine, above the 77–83 tok/s its paper reports on an RTX 5090, so our setup does not handicap it; FreeToken’s fetch cap and thread count were left at their defaults, while our FETCH table was swept. Our lead is therefore model-dependent: largest with few large experts (gpt-oss, top-4 of 128), smallest with many small ones.

Choosing the split per machine. On listing B, the law’s table beats the fixed table we used before at every budget of both models, by 3.3–8.0% (launch 1). Table 4 shows why at one budget: a single missed expert should run on the CPU (about $155\mu\text{s}$, hidden behind the GPU’s other experts) rather than be copied (about $225\mu\text{s}$, on the critical path); that entry alone is worth 3.7%. Fetching everything, FreeToken’s offload strategy expressed in our engine, is the worst table, and running everything on the CPU the second worst. The law’s table was within 1% of the best of five alternatives on launch 1 and tied with it on launch 2. Given each run’s measured misses, Eq. (3) ranks the tables correctly except law against lite, which are within 1%, and overstates their differences about $1.8\times$.

Machines with half the link. On RTX 5090s whose links we measured at 27–30 GB/s instead of 53 (jobs 085, 085b and 088: gpt-oss next to a Core i9-14900K, Qwen3 next to a Ryzen 9 7900 in a PCIe 4.0 slot), the law’s tables fetch less (gpt-oss 0,0,0,1,1, Qwen3 0,0,1,1,2,2,3,3,4) and beat the fixed table by 21–34% on gpt-oss and 9–18% on Qwen3, about twice the gain on listing B for Qwen3 and four to six times for gpt-oss: the per-machine split matters most where the link is slow. On the Ryzen, ours leads FreeToken by 1.04 – $1.08\times$; with the fixed table it would trail at 12.5%. On the Core i9 the first comparison (job 085) was confounded: FreeToken’s hybrid backend ran gpt-oss at 12–66 tok/s, below its own offload backend (25–86), because its CPU executor spread 23 threads across the 8 performance and 16 efficiency cores and its calibration then sent two thirds of the misses over the slow link, the same failure Section 3 reports for llama.cpp; the 1.4 – $2.0\times$ lead that gave us mea-

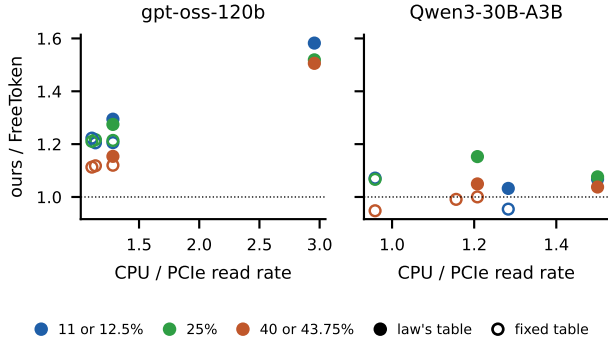


Figure 3: Ours over FreeToken’s better backend in every same-machine comparison of the current engine (jobs 076–088), against the machine’s CPU-to-PCIe read ratio (FreeToken’s own probe). Filled: FETCH table from the law on that machine; hollow: the fixed table. The rightmost gpt-oss points are the Core i9-14900K with FreeToken’s executor on its performance cores (job 088; the confounded job 085 points are omitted); the rightmost Qwen3 points are the Ryzen 9 7900.

sured a thread configuration. We reran it (job 088, a second rental of the same CPU model) with FreeToken’s executor on 8 threads pinned to the performance cores: its hybrid backend then runs $1.4\text{--}3.1\times$ faster than before (37.5, 62.1 and 92.5 tok/s; its calibration now copies 51.7% of misses), beating its own offload backend at 11 and 25% and tying it at 40%, and ours leads its better configuration by 1.58 , 1.52 and $1.51\times$ (lowest interval bound 1.47). Restricted to the CPU alone, FreeToken’s 8-thread executor decodes 13.7 tok/s at every budget. llama.cpp on the same rental runs 40.0 tok/s with one thread per performance core (-9.4% from the law’s prediction) and 32.8 with all 24 threads ($+25.9\%$): the hybrid-core failure is a thread-count choice, and the law holds once the choice is right. The law predicted llama.cpp on the Ryzen within $+2.7\%$. Figure 3 puts every same-machine comparison on one axis.

Long outputs. Our lead does not come from the first 256 tokens. On gpt-oss, on ten held-out hard problems (AIME 2022, problems 11–15 of both exams) that neither system has finished reasoning about after 2,048 tokens (at most 1 of 10 reach an answer), ours runs 1.36 [1.34, 1.38], 1.32 [1.30, 1.34] and 1.19 [1.17, 1.21] times FreeToken at 11, 25 and 40% over all 2,048 tokens, and 1.37 [1.34, 1.39], 1.33 [1.31, 1.35] and 1.19 [1.17, 1.22] over tokens 257–2,048 (job 083, listing A); both engines stream one event per token. On Qwen3, over tokens 257–2,048 of ten held-out MATH-500 problems, ours runs 1.043 [1.020, 1.070], 1.056 [1.036, 1.077] and 1.054 [1.025, 1.086] times FreeToken at 12.5, 25 and 43.75%. Both leads are larger than over the same runs’ first 256 tokens. Turning off FreeToken’s prefill overlap, which lends it cache slots during each prompt, changes its overall speed by 1% or less.

Table 5: Host reads per token relative to the exact optimum (Belady’s MIN with bypass over the whole future), median over nine models at three budgets, and the range over all 27 model–budget cells. Event-atomic replay of each model’s own sampled text; every online policy is scored on the last 90% of the trace. Mixtral (8 experts) uses $C = 2, 3$ and 4.

Policy	$C=E/8$	$E/4$	$3E/8$	range
LRU	1.44	1.77	2.03	1.37–2.26
LFU	1.44	1.93	2.40	1.37–6.15
Decayed frequency	1.45	1.70	1.94	1.35–2.10
with hysteresis κ	1.46	1.75	2.00	1.36–2.55
ARC	1.44	1.76	1.99	1.37–2.23
S3-FIFO	1.44	1.69	1.94	1.38–2.11
Static, profiled	1.92	2.74	3.48	1.43–13.24
Belady, $W=1$	1.18	1.45	1.69	1.07–2.06
Belady, $W=4$	1.01	1.13	1.31	1.00–1.95
Belady, $W=16$	1.00	1.00	1.01	1.00–1.66

Other systems and outputs. Two newer llama.cpp-based systems, Pipelined Sharding [37] and leloch’s expert cache [20], run gpt-oss-120b at $1.28\text{--}1.44\times$ stock llama.cpp; ours ran $1.42\text{--}2.34\times$ the better of them (job 077). Teacher-forced on fixed text (1536 steps per model) against stock llama.cpp with every weight in the VRAM of an RTX PRO 6000 (job 090), our cache at 25% has a mean KL divergence over the full vocabulary of 0.0019 nats on gpt-oss and 0.0005 on Qwen3 (99.9th percentile 0.121 and 0.059, maximum 0.21 and 0.13), picks the same next token on 98.5% and 99.3% of steps, and changes the loss by $+0.16\%$ and $+0.19\%$. Stock llama.cpp’s own `--n-cpu-moe` placement diverges from the same reference by as much (mean KL 0.0017 and 0.0008, agreement 98.8% and 99.3%): the differences are those of running experts on the CPU (the helpers quantise activations to Q8.0 where the GPU path uses Q8.1), not of the cache. Task accuracy was not measured.

6 What the Traces Say

The accounting names foresight as the largest cost where the host binds. What would recover it, and what would not? We replay the models’ own sampled text for 9 MoE models [27, 43, 28] (9 traces of 33,000–152,000 tokens, the cache carried across conversations) through an event-atomic simulator: each token requests its k experts per layer as one set, and no expert of the current token is evicted to serve it, the accounting that cache-level studies of this problem call for [49].

Policies. Table 5 scores LRU, LFU, decayed frequency, ARC, S3-FIFO and a profiled static cache against the exact optimum and against Belady within a window of W

future tokens, at three budgets per model. Three things hold across models. First, the online policies are close to one another and far from the optimum: the best of them reads 35–110% more than it, while decayed frequency is the best online policy in 12 of 27 cells and S3-FIFO in 10 (within 1% of the best in 20 and 18). Where the cache turns over every token ($C/k \leq 1.5$, 7 cells) the six online policies are within 4% of one another; from $C/k=2$ LRU reads 2–11% more than decayed frequency on eight of nine models, Qwen2-57B being the exception, and classic LFU fails at large budgets ($6.2\times$ the optimum). The literature’s disagreement on recency against frequency [33, 19, 42, 11] is consistent with this regime dependence. Second, a static cache profiled on other text reads $1.4\text{--}13.2\times$ the optimum, because the hot set moves: the C most requested experts of a layer change by 22–58% between consecutive windows of 100 tokens and 33–75% between windows of 1,000 (median 51%), more over the longer window on 8 of nine models, so popularity follows the conversation; the skew is mild, the top quarter of experts taking 31–56% of requests. Third, four tokens of foresight beat every online policy in every cell (median $1.13\times$ the optimum at $E/4$), so foresight is worth several times the choice of policy. Our deployed hysteresis costs 1–8% reads over plain decayed frequency (up to 22% on Qwen3); it was tuned for time, where it rations copies, not for reads.

How far ahead. The optimum reads 26–56% fewer experts from host memory than the best online policy (median 41%). The lookahead needed scales with the cache’s turnover time C/k , slots per layer over experts per token: half of the gap closes at $W_{50} \approx 0.59 (C/k)^{1.33}$ tokens ($r=0.971$ in log-log over 26 points; the exponent’s 95% interval over a bootstrap of models is 1.09–1.47, its leave-one-model-out range 1.24–1.37, and without the 6 points whose W_{50} lies below one token it is 1.51), and 90% needs a median of $2.7 C/k$. Small caches ($C/k \leq 2$) recover half of the gap with one or two tokens of foresight; gpt-oss-120b at 25% ($C/k=8$) needs about 10 tokens, and at 50% about 34. The law adds the condition: foresight is worth these reads only where the host binds; where the bus is already saturated, as on the slow devices of [47, 46], it is worth nothing. Predicting the next layer’s experts, the usual form of prediction [15, 52], gives no cross-token foresight: it can raise the hit rate by prefetching but never lowers host bytes, which is what we measured when we built it.

Speculation is not foresight. Speculative decoding verifies K draft tokens in one forward pass, so a layer’s experts for K positions are requested together and a missed expert is read once for all of them, which several systems propose as the source of foresight for a cache [6, 22, 30]. On the same traces, with K positions per step and drafts accepted at rate α until the first rejection, batched verification saves an online cache a median 15–32% of its reads across models at $K=8$ when rejected drafts cost nothing

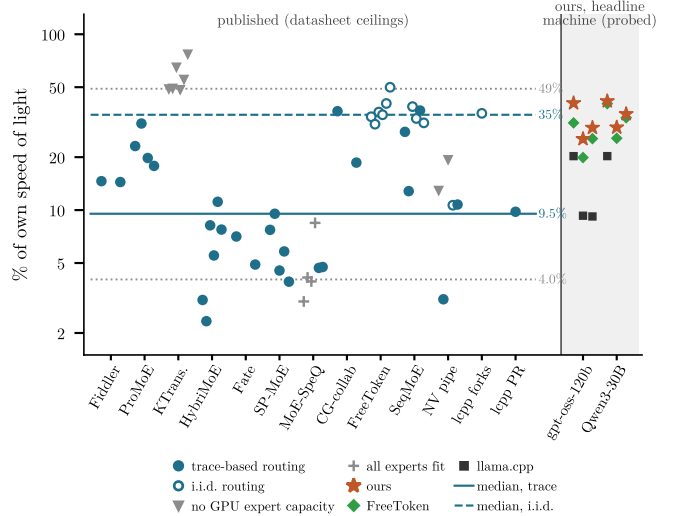


Figure 4: Published MoE-offloading measurements (52 rows from 13 sources) as a percentage of their own speed of light, computed from the paper’s hardware at datasheet rates and, where a routing trace of the model exists, the per-layer Belady optimum (filled); hollow points assume uniform routing, which overstates the optimum’s reads and so the percentage. Right: our cache, FreeToken and stock llama.cpp on listing B against the probed limit of Table 1.

(48% on the model that gains most), the equivalent of a window of two to four tokens; once the rejected drafts’ experts are counted, the saving is 8–12% at $\alpha=0.9$ and becomes a loss of 18–20% at $\alpha=0.8$, because the union of K positions’ expert sets is 20–79% of Kk , several times C (Appendix B lists the variants). The optimum gains little from batching (–7 to 15%). Foresight must come from knowing future routing, as a router decoupled from the backbone [3] or a forecaster [39] would provide, and W_{50} says how far ahead.

7 Published Systems Against Their Own Limit

Rule 11 of scientific benchmarking asks for an upper bound beside every measurement [13]. We applied a bound of the same construction as Eq. (2) (Belady’s optimum with bypass per layer, CPU and PCIe paths, a maximum per layer step; `scripts/audit.py`), with each paper’s stated hardware at datasheet rates and a routing trace of its model where we have one, to 52 published measurements of MoE offloading from 13 sources, collected for an earlier audit of this literature (Appendix C). Figure 4 shows the result. The 29 trace-based measurements reach a median 9.5% of their own speed of light (quartiles 4.9–17.8%, best 36.8%); the 11 that needed an i.i.d. routing assumption, 34.8%, an overestimate; measurements with no GPU expert capacity (the CPU-only rate is their bound) and with

every expert resident are plotted apart. Our own reference points are not like for like: their ceilings are probed, and a probed ceiling is lower than a datasheet one by the ratio of achieved to nominal host bandwidth, which we did not record for the published machines and so do not quantify; it would raise the published percentages by that ratio. Even so, the trace-scored measurements sit at a median tenth, and three quarters of them below a fifth, of what their machines could deliver, and Fig. 1 says what the missing part is made of.

8 Related Work

Offloading systems. Demand-paging systems cache experts per layer and copy misses over PCIe [8, 41, 34, 36, 31]. CPU-execution systems run non-resident experts on the host [18, 5, 51, 50, 38]; a set-associative GPU cache with CPU-executed misses [14], a llama.cpp fork [20, 21] and two llama.cpp pull requests [26, 7] are the closest designs to ours, and FreeToken [44] and DALI [53] split each layer’s misses between CPU and PCIe, FreeToken from a closed-form bandwidth model. Pipelined Sharding [37] plans sub-layer placement under a VRAM budget. Predictors guess upcoming experts from the previous layer’s state [15, 52], several steps ahead with a learned forecaster [39, 42], or a batch’s routes across layers [10]; ReadME [3] makes routing known in advance by decoupling the router and then caches in Belady’s manner. Speculative decoding has been combined with expert offloading as a source of batching [6, 22, 30]. Batched-throughput systems [32, 4, 45] and SSD tiers [24, 25] are outside our single-request, host-DRAM scope.

Models, bounds and audits. FreeToken’s split and our Eq. (3) are bandwidth models of one layer; Eq. (1) is a whole-token model tested blind across machines and against alternatives. Max-of-paths cost models go back to FlexGen [32] and MoE-Lightning [4]; WiSP [47] bounds decode by PCIe turnover, Budgeting Bytes [46] by bytes over bandwidth, and MoE-CAP [17] measures bandwidth utilisation against a single peak. Qiu et al. [29] predict llama.cpp throughput from GGUF metadata for models held entirely in GPU or unified memory; Suram [35] matches one offloaded configuration to bytes per token over bandwidth; Si et al. [33] derive when a copy pays off and find recency competitive with frequency at large budgets. The roofline [40] is the single-memory ancestor of our bound. Zhang [49] proposes a trace-level evaluation contract for expert caching, decomposes the online-to-optimal gap in cache blocks in one fixed order, and audits the reporting of ten papers; SeqMoE [39] reports four systems against the all-in-VRAM speed. Trace studies of policies cover two or three models each [49, 33, 19, 11]. Shapley attribution of a performance gap has a precedent in ShapleyIQ [23].

Lookahead caching. Limited lookahead in paging goes back to Albers [1]; Hasslinger et al. [9] combine Belady within a window with LRU or LFU beyond it for web caches, the policy family we use to measure the horizon.

9 Limitations

- **Hardware.** One GPU model (RTX 5090; an RTX PRO 6000 for all-in-VRAM runs) and desktop Ryzen and Intel hosts, rented; GPU clocks could not be locked, power limits differed (400–575 W), and listing B’s memory runs $1.21\times$ faster than the other cards’. A 12-channel server CPU breaks the law’s bandwidth assumption (Section 3). Intervals cover problem sampling on one machine, not machine-to-machine variance.
- **Workload.** Single-request decode; AIME prompts, which were also used while developing the system; greedy decoding; 130–186-token prompts. Long outputs are measured on held-out problems. Prompts, batches above one and speculative verification bypass the cache and are not measured.
- **Competitor settings.** FreeToken ran its faster backend per budget with its own bandwidth calibration, default thread count and fetch cap, while our FETCH table was swept; its CPU backend failed on a hybrid-core host at its default thread count and was rerun on the performance cores. Each comparison uses one confirmation launch, except Qwen3 at 25%, which is from the selection launch for both systems.
- **The bound and the audit.** Exact routing only: expert skipping, substitution or deferral change the model and fall outside it. The published rows use datasheet ceilings and, for eleven of them, i.i.d. routing.
- **Constants and outputs.** G , g and L_c are fitted once and frozen; the per-layer model overstates differences between FETCH tables about $1.8\times$. Output parity is measured by teacher-forced KL divergence, agreement and loss against an all-VRAM reference, not by task accuracy.

10 Conclusion

On a consumer machine, MoE decode time is a fixed GPU term plus host-memory bytes over the rate of the paths that read them. That law, predicted blind and tested against its alternatives, is accurate to a few percent on desktop hosts; it tells a system which path to use for a miss on a given machine; and with the optimum’s bytes it gives a speed limit in seconds that published systems reach a median tenth of. An accounting that no order of fixes can bias says what the rest is: where the host binds, the policy’s ignorance of future routing, and where the GPU

binds, host reads that wait for it. Nine models’ traces say that neither an online policy nor batching alone buys that foresight; it must come from knowing the routing a few to tens of tokens ahead.

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A Scientific benchmarking checklist

How this paper follows the twelve rules of Hoefer and Belli [13].

Rule	Where and how
1. Report the base case of a speed-up and its absolute performance.	Every ratio in Table 1 is next to both absolute speeds; llama.cpp is configured with per-layer expert offload at equal expert memory, the setting its users run.
2. Explain subsets of benchmarks or resources.	Two models chosen for size and precision (MXFP4, BF16), a third that is FreeToken’s own headline model; budgets span what fits a 32 GB card; the 43.75% budget is omitted where it does not fit. All 30 AIME-25 problems are used.
3. Arithmetic mean for costs, harmonic mean for rates.	Every problem decodes the same number of tokens, so the harmonic mean of rates is total tokens over total time; ratios of total time agree with the reported ratios of mean rates within 0.003.
4. Avoid summarising ratios.	Ratios are computed from mean speeds, never averaged across budgets or models; ranges list the per-budget values.
5. Report whether measurements are deterministic; give confidence intervals.	Decode is greedy and deterministic per engine; timing is not. Every comparison has a 95% interval.
6. Do not assume normality.	Intervals are bootstrap percentile intervals (10,000 resamples, paired by problem); no distributional assumption.
7. Compare nondeterministic data soundly.	Paired bootstrap of the ratio of means; the variant a system is scored with is picked on a separate launch (except the Qwen3 25% row of Table 1).
8. Check that central tendency is the right summary.	Mean speed over 256 tokens is FreeToken’s own metric; per-token p50 and p99 latencies are recorded for every problem in the released rows.
9. Document all factors and the setup.	Machines, probes, power limits, memory clocks (Section 2), software commits (llama.cpp 4da6337, FreeToken 0d652e7), flags and job scripts are released; Appendix C.
10. Report measurement, synchronisation and summarisation.	One client times every system from streamed token arrivals; no server-side timers enter a comparison (Section 2).
11. Show upper performance bounds.	The speed limit (Section 4) and all-in-VRAM speed are columns of Table 1; Table 3 tightens the limit four ways; Section 7 applies it to the published literature.
12. Plot enough to interpret; connect points only for trends.	Figures 1 and 2 are bar charts; Figs. 3 and 4 are scatter plots; no interpolation.

B Prediction log

Each job’s predictions are in the header of its script on the public `gpu` branch, committed and pushed before the machine started; git history orders every prediction before its measurement. Outcome notes with the statistics are in `prereg/`. Table 7, generated from `prereg/scorecard.clauses.json`, splits every prediction of jobs 073–087 into its separable clauses and scores each under one rule: *held* only when the point estimate is on the predicted side and the 95% paired-bootstrap interval excludes the threshold (for a band, lies inside it); *held (point)* when only the point estimate is, or no interval exists; *failed* when the point estimate is on the wrong side; *untested* when the clause could not be evaluated; *void* when the job was voided. Of 97 clauses, 55 held, 14 held on the point estimate, 22 failed, 2 were untested and 4 void: 7 of 19 scored clauses held in jobs 073–075, when the system was being built, 19 of 23 in jobs 076–081, 11 of them sign-only predictions made after a same-host pilot, 24 of 38 in jobs 082–087, and 5 of 11 in the jobs run after the reviews (088 onward; the three failures there are one band, our slow-link lead landing at 1.51–1.58× against a predicted 1.10–1.50). Against the earlier draft’s log, two clauses are downgraded (a llama.cpp clause of job 074b that the log counted as held was never run; job 085b’s own restatement of its fourth prediction, “at least 2 of 3”, failed at 1 of 3, though the pooled 4 of 6 held), job 073’s “identical outputs” is scored on fully identical outputs (0–3%) rather than on agreement of the first 160 characters (27–47%), and ten clauses the log counted as held, among them job 087’s first, hold on the point estimate only under the interval rule; `prereg/scorecard.outcome.md` lists every change. Job 085’s Qwen3 half ran as 085b on a second machine with a slow link after a hung conversion. Job 084 ran

on a throttled card and was rerun, with its predictions unchanged, as 084b/084c. The batched-verification variants of Section 6 (rejected drafts free, routed like the true token, or routed at random) are in `prereg/batchk_outcome.md`.

Table 7: Prediction scorecard, jobs 073–092: every committed prediction split into its separable clauses and scored under one rule. *Held*: the point estimate is on the predicted side and the 95% paired-bootstrap interval excludes the threshold (for a band, lies inside it). *Held (point)*: the point estimate is on the predicted side but the interval includes the threshold, or no interval exists. *Untested*: the clause could not be evaluated. *Void*: the job was voided (084, a throttled card; its predictions are scored on 084b). Types: sign (a direction), threshold (a numeric bar), band (an interval), equality (a deterministic outcome). Measured values are ratios of mean speeds with 95% intervals unless the clause says otherwise; `prereg/scorecard_clauses.json` names the source of every number. Totals: 97 clauses, 55 held, 14 held (point), 22 failed, 2 untested, 4 void.

Job	Clause	Type	Measured	Status
Jobs 073–075: 20 clauses, 7 held, 3 held (point), 9 failed, 1 untested, 0 void				
073 P1a	ours+FETCH $\geq$ 5% ahead of FreeToken, 11%	threshold	1.109 [1.098, 1.120]	held
073 P1b	ours+FETCH $\geq$ 5% ahead of FreeToken, 25%	threshold	1.084 [1.076, 1.092]	held
073 P1c	ours+FETCH within $\pm 5\%$ of FreeToken, 40%	band	0.941 [0.933, 0.951]	failed
073 P2	ours+FETCH $\geq 1.8\times$ llama.cpp, every budget	threshold	1.832 [1.808, 1.856]; 2.358 [2.322, 2.396]; 2.647 [2.605, 2.686]	held
073 P3a	same-problem warm-up reads 5-15% faster, the two caches	band	0.991 [0.962, 1.028]; 0.997 [0.986, 1.008]	failed
073 P3b	same-problem warm-up within $\pm 2\%$, llama.cpp	band	0.962 [0.934, 0.979]	failed
073 P4	$\geq 80\%$ of greedy outputs identical to llama.cpp’s, every system	equality	ours (6 configurations): 0.000-0.033; FreeToken (6 configurations): 0.000; llama.cpp -ncmoe 32 / 22: 0.700 / 0.400	failed
074 P1	ec-bench launch time ≤ 0.25 ms, C51	threshold	0.201	held (point)
074 P2a	ours v2 ≥ 127 tok/s, C51	threshold	95.1	failed
074 P2b	ours v2 ahead of FreeToken offload, C51	sign	0.831 [0.817, 0.845]	failed
074 P3a	ours v2 ≥ 59 tok/s, C14	threshold	41.5	failed
074 P3b	ours v2 ≥ 91 tok/s, C32	threshold	65.3	failed
074 P4	half-life 64 within $\pm 2\%$ of half-life 16, C51	band	1.012 [0.997, 1.027]	held (point)
074b P1	GPU sampling cuts between-step host time by ≥ 1 ms	threshold	1.6	held (point)
074b P2a	ours v2 $\geq 10\%$ faster with GPU sampling, C51	threshold	1.117 [1.109, 1.125]	held
074b P2b	stock llama.cpp $\geq 3\%$ faster with GPU sampling	threshold	–	untested
074b P3	ours v2+FETCH with GPU sampling $\geq$ FreeToken offload, C51	sign	1.074 [1.063, 1.085]	held
075 P1	ours ahead of FreeToken at 40% (RTX PRO 6000)	sign	1.122 [1.107, 1.137]	held
075 P2	ours within -3 to +10% of FreeToken at 60%	band	1.118 [1.109, 1.127]	failed
075 P3	ours at 60% $\geq 55\%$ of the all-in-VRAM speed	threshold	0.726 [0.717, 0.735]	held
Jobs 076–081: 23 clauses, 19 held, 4 held (point), 0 failed, 0 untested, 0 void				
076 P1	ours ahead of FreeToken at 11/25/40%	sign	1.205 [1.190, 1.220]; 1.217 [1.201, 1.234]; 1.118 [1.108, 1.129]	held
076 P2	ours $\geq 1.8\times$ llama.cpp, every budget	threshold	1.924 [1.894, 1.956]; 2.683 [2.631, 2.735]; 3.191 [3.130, 3.250]	held
076 P3	the law within 8% on own-text runs (four configurations)	band	1.009 [0.962, 1.066]; 1.006 [0.947, 1.086]; 1.052 [0.960, 1.164]; 0.960 [0.909, 1.024]	held (point)
077 P1a	both entrants run end to end on gpt-oss-120b	equality	30 of 30 problems at every budget for both	held
077 P1b	both entrants faster than llama.cpp at 25/40%	sign	pipeshard 25 / 40%: 1.348 [1.336, 1.360]; 1.417 [1.399, 1.435]; leloch 25 / 40%: 1.446 [1.431, 1.461]; 1.280 [1.269, 1.291]	held
077 P2	ours ahead of both entrants, every budget	sign	1.421 [1.403, 1.439]; 1.896 [1.871, 1.921]; 2.343 [2.293, 2.391]	held
078 P1	ours ahead of FreeToken at 12.5/25% (Qwen3)	sign	1.071 [1.063, 1.080]; 1.067 [1.043, 1.094]	held
078 P2	ours ahead of KTransformers, every budget	sign	2.168 [2.140, 2.197]; 3.059 [2.975, 3.162]; 4.420 [4.310, 4.550]	held
078 P3	ours $\geq 1.5\times$ llama.cpp, every budget	threshold	2.122 [2.092, 2.151]; 3.051 [2.967, 3.152]; 4.470 [4.350, 4.609]	held
079 P1a	host-side time per step < 1 ms	threshold	0.266	held (point)
079 P1b	host-side time $< 10\%$ of the token time	threshold	0.027	held (point)
079 P2	$\geq 1.5\times$ FreeToken’s kernels per token	threshold	1.606	held (point)
079 P3	ours within $\pm 5\%$ of FreeToken at 43.75%	band	0.991 [0.975, 1.009]	held
080 P1	law table $\geq 3\%$ over the current table, 43.75%	threshold	1.048 [1.041, 1.055]	held
080 P2	law table within 2% of the sweep’s best, 43.75%	band	0.991 [0.986, 0.996]	held

continued

Job	Clause	Type	Measured	Status
080 P3	best table leads FreeToken, 43.75%	sign	1.049 [1.036, 1.064]	held
080 P4	law table beats the current table, 25%	sign	1.067 [1.062, 1.071]	held
081 P1	law table beats current, gpt-oss, every budget	sign	1.079 [1.072, 1.085]; 1.054 [1.041, 1.066]; 1.033 [1.021, 1.044]	held
081 P2	ours leads FreeToken, gpt-oss 11/25/40%	sign	1.294 [1.278, 1.312]; 1.275 [1.254, 1.295]; 1.154 [1.134, 1.172]	held
081 P3a	law table beats current, Qwen3 12.5%	sign	1.080 [1.073, 1.084]	held
081 P3b	ours leads FreeToken, Qwen3 12.5%	sign	1.032 [1.022, 1.042]	held
081 P4a	ours $\geq 1.8\times$ llama.cpp, gpt-oss, every budget	threshold	2.001 [1.964, 2.041]; 2.731 [2.668, 2.795]; 3.196 [3.133, 3.259]	held
081 P4b	ours $\geq 2\times$ llama.cpp, Qwen3 12.5%	threshold	2.051 [2.026, 2.077]	held
Jobs 082–087: 43 clauses, 24 held, 4 held (point), 10 failed, 1 untested, 4 void				
082 P1	steady-state lead over FreeToken, gpt-oss, held-out prompts	sign	–	untested
082 P2	FreeToken gains more after 256 tokens than ours, every budget	sign	FreeToken: 1.017 / 1.065 / 1.071; ours: 1.038 / 1.094 / 1.117	failed
082 P3	FreeToken ahead in steady state, Qwen3 43.75%	sign	1.054 [1.025, 1.086]	failed
082 P4a	every ablation step $\geq -1\%$	threshold	0.723 [0.712, 0.733]; 0.744 [0.740, 0.748]	failed
082 P4b	ours $\geq 2\times$ stock at 25%, both models	threshold	2.658 [2.566, 2.749]; 2.800 [2.743, 2.858]	held
082 P5a	parity: $\geq 98\%$ top-1 agreement, both models	threshold	0.986 / 0.993	held
082 P5b	parity: $ \Delta\text{NLL} \leq 1\%$, both models	threshold	-0.0022 / 0.0031	held
082 P6	llama.cpp Qwen3 25/43.75% within 5% of the law	band	0.966 [0.962, 0.969]; 0.962 [0.958, 0.965]	held
083 P1	still reasoning at 2,048 tokens on ≥ 8 of 10, both systems	threshold	ours: 9 / 9 / 10; FreeToken: 10 / 9 / 10	held
083 P2	ours within 15% of its Table 1 rate	band	1.046 [1.019, 1.072]; 1.037 [1.013, 1.060]; 1.033 [1.003, 1.061]	held
083 P3a	ours leads FreeToken, all tokens, 11/25/40%	sign	1.358 [1.338, 1.378]; 1.322 [1.303, 1.342]; 1.191 [1.172, 1.212]	held
083 P3b	ours leads FreeToken, tokens 257-2,048	sign	1.366 [1.343, 1.389]; 1.328 [1.310, 1.350]; 1.194 [1.175, 1.215]	held
083 P4	lead $\geq 15\%$ at 11 and 25%	threshold	1.358 [1.338, 1.378]; 1.322 [1.303, 1.342]	held
084 P1	Qwen3 all in VRAM at 120-190 tok/s	band	–	void
084 P2	gpt-oss all in VRAM within 5% of 261.4 tok/s	band	–	void
084 P3	ours with 128 slots within 5% of all in VRAM, both	band	–	void
084 P4	sim. hit rate within 5 points; ours at 20-50% of the limit	band	–	void
084b P1	Qwen3 all in VRAM at 120-190 tok/s	band	177.8 [177.1, 178.4]	held
084b P2	gpt-oss all in VRAM within 5% of 261.4 tok/s	band	0.977 [0.971, 0.982]	held
084b P3	ours with 128 slots within 5% of all in VRAM, both	band	0.997 [0.991, 1.004]; 1.047 [1.043, 1.052]	held (point)
084b P4a	sim. hit rate within 5 points of the engine’s, every budget	band	-0.014 / -0.008 / -0.0003 / -0.053 / -0.026 / -0.006	failed
084b P4b	ours at 20-50% of the speed limit, every budget	band	0.413 / 0.256 / 0.294 / 0.433 / 0.302 / 0.352	held (point)
084c	<i>No predictions of its own: it supplies the gpt-oss routing trace that 084b-P4a and 084b-P4b are scored on.</i>	–	–	–
085 P1	law’s tables fetch less than on the headline machine	equality	gpt-oss: 0,0,0,1,1 (two entries lower); Qwen3: 0,0,0,1,1,2,2,3,3 (six entries lower)	held
085 P2a	law beats fixed, gpt-oss, every budget	sign	1.340 [1.327, 1.355]; 1.274 [1.256, 1.292]; 1.211 [1.187, 1.235]	held
085 P2b	law’s gain larger than headline at ≥ 2 of 3 (pro rata of 4 of 6)	threshold	3 of 3 (every interval excludes the headline value); pooled with 085b: 6 of 6	held
085 P3	ours leads FreeToken, gpt-oss, every budget	sign	2.030 [1.993, 2.068]; 1.699 [1.662, 1.737]; 1.428 [1.391, 1.463]	held
085 P4	lead larger than headline at ≥ 2 of 3 (pro rata of 4 of 6)	threshold	3 of 3 (every interval excludes the headline value); pooled with 085b (1 of 3): 4 of 6	held
085 P5	llama.cpp at 25% within 5% of the law, gpt-oss	band	0.818 [0.817, 0.819]	failed
085b P1	law’s table fetches less than on the headline machine	equality	0,0,1,1,2,2,3,3,4 (three entries lower)	held
085b P2a	law beats fixed, Qwen3, every budget	sign	1.184 [1.177, 1.190]; 1.149 [1.142, 1.157]; 1.094 [1.086, 1.101]	held
085b P2b	law’s gain larger than headline at ≥ 2 of 3	threshold	3 of 3 (every interval excludes the headline value)	held
085b P3	ours leads FreeToken, Qwen3, every budget	sign	1.067 [1.058, 1.076]; 1.076 [1.063, 1.089]; 1.038 [1.017, 1.060]	held

continued

Job	Clause	Type	Measured	Status
085b P4	lead larger than headline at ≥ 2 of 3	threshold	1 of 3: 12.5% 1.067 [1.058, 1.076] > 1.032 ; 25% 1.076 [1.063, 1.089] < 1.153 ; 43.75% 1.038 [1.017, 1.060] below 1.049 with the interval covering it	failed
085b P5	llama.cpp at 25% within 5% of the law, Qwen3	band	0.974 [0.972, 0.976]	held
086 P1	FreeToken as shipped within 15% of its paper's rate	band	93.8 [91.3, 96.5]	held (point)
086 P2	ours leads FreeToken at 12.5/25/37.5%	sign	1.030 [1.023, 1.038]; 1.010 [0.998, 1.021]; 1.003 [0.994, 1.013]	failed
086 P3	ours $\geq 2\times$ llama.cpp, every budget	threshold	1.976 [1.950, 2.005]; 2.311 [2.264, 2.361]; 2.481 [2.441, 2.521]	failed
086 P4	law beats fixed, every budget	sign	1.071 [1.068, 1.074]; 1.060 [1.057, 1.063]; 1.057 [1.052, 1.061]	held
087 P1	profiled static cache $\geq 1.3\times$ stock, both models	threshold	1.307 [1.236, 1.379]; 1.889 [1.803, 1.969]	held (point)
087 P2	replacement adds less than placement, both models	sign	0.268 [0.122, 0.434]; -0.765 [-0.898, -0.619]	failed
087 P3	decayed frequency beats the profiled static cache, both	sign	1.575 [1.497, 1.674]; 1.124 [1.070, 1.187]	held
087 P4a	every step after the static cache $\geq -1\%$	threshold	0.877 [0.835, 0.924]	failed
087 P4b	ours $\geq 2\times$ stock, both models	threshold	2.680 [2.595, 2.763]; 2.815 [2.754, 2.877]	held
Jobs 088–092: 11 clauses, 5 held, 3 held (point), 3 failed, 0 untested, 0 void				
088 P1	FreeToken hybrid on 8 threads $\geq 1.3\times$ job 085's hybrid, every budget	threshold	3.12 / 2.04 / 1.4	held (point)
088 P2	FreeToken's best improves on job 085's best at ≥ 2 of 3 budgets	threshold	1.52 / 1.29 / 1.08	held (point)
088 P3a	ours leads FreeToken's best, 11% (CI > 1)	sign	1.582 [1.564, 1.603]	held
088 P3b	ours leads FreeToken's best, 25% (CI > 1)	sign	1.519 [1.472, 1.559]	held
088 P3c	ours leads FreeToken's best, 40% (CI > 1)	sign	1.506 [1.476, 1.537]	held
088 P3d	ours / FreeToken between $1.10\times$ and $1.50\times$, 11%	band	1.582 [1.564, 1.603]	failed
088 P3e	ours / FreeToken between $1.10\times$ and $1.50\times$, 25%	band	1.519 [1.472, 1.559]	failed
088 P3f	ours / FreeToken between $1.10\times$ and $1.50\times$, 40%	band	1.506 [1.476, 1.537]	failed
088 P4a	llama.cpp -t 8 within 10% of the law	band	-9.4	held (point)
088 P4b	llama.cpp -t 8 faster than -t 24	sign	1.219 [1.213, 1.225]	held
088 P5	the law's table is 0,0,0,1,1 or fetches less	equality	0,0,0,1,1	held

C Artifact description

- **System.** A patch to llama.cpp commit 4da6337 (the expert cache, FETCH, the mailbox and the llama-ec-bench instrumentation: teacher-forced loss, logit dumps, routing lookahead records); built with CUDA 13.0 for compute capability 12.0.
- **Jobs.** One shell script per job; each installs FreeToken at commit 0d652e7, downloads the models from Hugging Face, runs the bandwidth probes, writes any on-machine prediction, runs the client, and stores raw rows (one JSON line per problem), server logs and engine counters. A small runner rents a Vast instance, streams a heartbeat, fetches the results and only then destroys the instance.
- **Analysis.** scripts/*_stats.py compute every interval from the raw rows; scripts/perlayer_model.py, speed_limit_v2.py, shapley_gap.py, policy_study.py, foresight_exact.py, batchk_trace.py, fig_audit_sol.py and scorecard.py produce the model comparison, the limit, the accounting, the trace studies, the audit figure and the scorecard; scripts/wsg_tables.py and wsg_numbers2.py generate every table, figure and number in this paper from their outputs. The audit's 52 published rows and the physical speed-of-light model they are scored against are data/published_measurements.json, data/audit/ and scripts/audit.py.
- **Cost.** The 46 rentals (26 distinct machines) of jobs 058–087 cost 29.6 US dollars at their hourly GPU prices, by the ledger in the repository; storage and download charges, which the ledger omits, add to this.